

ELHOUSSAINE GANGOUCH

+212 682 783 853 | elhousaainegangouch@gmail.com |

 [elhousaainegangouch](#) |  [GANGOUCH1](#) |

Benguerir, Morocco

PROFESSIONAL SUMMARY

Research Engineer | Applied Math Operational Research Specialist in high-efficiency algorithms for LLM optimization and RAG/RAFT systems. Published researcher (IJICCA) with a focus on mathematical modeling to accelerate inference and reduce enterprise costs. Expertise in transforming theoretical innovation into scalable, production-grade AI solutions

EXPERIENCE

- **Data Scientist / Research Engineer** February 2025 - February 2026
UM6P College of Computing Benguerir, Morocco
 - Designed and productionized deep learning architectures for multivariate time series forecasting, Developed **SPECTRA**, a novel framework for Retrieval-Augmented Forecasting (RAFT), achieving a 100x reduction in memory costs and 10x inference acceleration for industrial-scale deployment.
 - Optimized LLM performance through holistic low-rank compression (SVD), directly contributing to cost governance and reduced latency for enterprise-grade AI deployment.
 - Engineered hybrid neural architecture by integrating Isogeometric Analysis (IGA) basis functions into Kolmogorov-Arnold Networks (KAN-IGA), enabling structured, smooth activations for time series forecasting.
 - Architected and deployed a **full-stack AI platform** using **FastAPI** and **React**, implementing RESTful APIs to bridge backend LLM logic with interactive frontend interfaces.
- **Research Intern - Numerical Approximation** March 2024 – Sept 2024
UM6P Vanguard Center Benguerir, Morocco
 - Developed CAD-integrated code to solve Maxwell's equations using finite element methods and de Rham complex sequences for structure-preserving discretization.
 - Scaled and preconditioned large-scale linear systems to improve solver convergence for high-performance scientific computing tasks.
- **Technical Contributor – AMLD Africa 2026** January 2026
1337AI, OCP Solutions, Data Science Africa Benguerir, Morocco
 - Contributed 30 hours of dedicated technical participation to the Applied Machine Learning Days (AMLD) summit, focusing on the democratization of AI across the African continent.
- **Mentor – NASA Space Apps Challenge** October 2025
University Mohammed VI Polytechnic (UM6P) Benguerir, Morocco
 - Guided multidisciplinary teams in developing ML and mathematical solutions for NASA Earth observation, acting as a technical lead to turn complex needs into data products.
 - Assisted teams in using AI and numerical simulation to prototype real-world applications for climate and remote sensing.

TECHNICAL SKILLS

- **Programming & Data:** Python (PyTorch, Hugging Face, NumPy, SciPy, Pandas), ETL, API Integration
- **Generative AI:** RAG / RAFT (Retrieval-Augmentation), Large Language Models (LLMs), Prompt Engineering, Model Compression (SVD, Pruning, Quantization), Attention Mechanisms
- **Scientific Computing:** Time Series Forecasting, Optimization, Mathematical Modeling, Operational Research, Hypothesis Testing, Regression Analysis, Isogeometric Analysis (IGA), Finite Element Method (FEM), PDE Discretization

EDUCATION

- **Faculty of Sciences Ain Chock** October 2021 - July 2024
Master's Degree in Numerical Methods and Applied Statistics Casablanca
 - Major: Numerical Methods and PDEs
 - Thesis: Numerical Approximation of Maxwell's Equations using IsoGeometric Analysis
- **Faculty of Sciences Ain Chock** October 2018 - July 2021
Bachelor in Applied Mathematics Casablanca
 - Major: Partial Differential Equations (PDEs)
 - Thesis: Wigner Transform and Fourier Transform

PUBLICATIONS

- Issam Aït Yahia, Elhoussaine Gangouch, Abdelkader El Mahdaouy, Ismail Berrada. **SPECTRA: Sparse Periodic Extraction for Efficient Retrieval-Augmented Forecasting**. Submitted to IJCAI-ECAI 2026.

CERTIFICATIONS

- Deep Learning Specialization (Coursera) October 2023
- Professional Certificate in Deep Learning (IBM) October 2022

ACADEMIC LEADERSHIP & OUTREACH

- Academic Mentorship Lead / Tutor March 2023 – January 2024
Faculty of Sciences Ain Chock Casablanca, Morocco
 - Built and managed a team of **six PhD student volunteers** to provide academic support and peer mentoring for 100+ first-year STEM students.
 - Designed and coordinated 20+ tutoring sessions (TD), focusing on foundational Mathematics and Physics to bridge the gap for incoming university students.
 - Developed comprehensive course reviews and exam preparation guides, significantly improving student performance in core science modules.

ACADEMIC ENGAGEMENT & PROFESSIONAL DEVELOPMENT

- Spring School - AI For Impact Selected participant in intensive program for social-impact AI applications.
UM6P, Benguerir March 2026
- Winter School: Next-Gen AI & Economic Applications Completed program on innovation policy and data-driven decision making.
UM6P, Benguerir March 2025
- HPC AI Days – Hackathon Participant Collaborated on hybrid CPU/GPU architecture challenges for matrix computations.
UM6P, Benguerir December 2024
- International eHealth Forum 2024 Attended sessions on biomedical AI and digital health policy for global systems.
UMSS, Casablanca October 2024

ADDITIONAL INFORMATION

Languages: Arabic (Native), Amazigh (Native), French (Advanced - B2), English (Advanced)